

Week 1: Jumping pegs Facilitator guide (W1-F-v1)

This guide is the only place with answers, arguments, timings and teaching notes. The student pages contain problems, boards and recording space and nothing else. Please read your own table's section fully before the session; every answer is worked out here so that nothing has to be solved at the table. Section 9 is background for the two mathematicians; section 10 is a page to hold up at the share.

1. The session at a glance

The move. A peg jumps over an orthogonally adjacent peg into the empty square just beyond it, and the jumped peg is removed. That is the only move all hour.

The questions. All three tables meet the same three questions in different sizes: *Can this pattern of pegs be cleared down to one peg? How far past a line can an army of pegs send a single peg? How few pegs does that take?* The headline facts: on a strip, an army of any size sends a peg exactly one square past the line, never two; on a grid, armies of 2, 4, 8 and 20 pegs reach the first, second, third and fourth row past the line, and no army, however large, reaches the fifth (Conway's Soldiers). The tool that proves the impossibilities is a value attached to each square so that no jump ever raises the total value of the pegs on the board. The K-1 table meets the facts; the grades 2-3 table proves the strip fact with values; the grades 4-5 table proves the strip fact and the grid facts.

0:00-0:05 Free handling. Pieces and page 1 are on the tables; children do whatever they like with them.

0:05-0:08 Whole-group launch (section 3): the jump, done by everyone on strip A.

0:08-0:50 Table work (sections 5-7). Roughly forty minutes.

0:50-0:58 Share (section 8): each table shows one thing, a child from another table says it back.

0:58-1:00 Pieces back in cups, packets go home.

Core and Extra. Each packet has a core sequence and, at the back, problems labelled *Extra*. A table that does only the core problems has done the whole session: it has met the move, the river, the army and the reason. The Extra pages are for a table that finishes; skip them without comment otherwise. Every section below says what to do at 0:40 if the table is behind, so that each table ends on its finale.

Who sits where. Parent at K-1 (section 5 starts with an answer box, and nothing in it has to be worked out), second mathematician at grades 2-3, organizer at grades 4-5. Every adult stays at their own table for the whole forty minutes; the K-1 section is written so that the parent needs nothing else. Swap the mathematicians if you prefer; both sections are self-contained.

2. Materials and preparation

- Student packets, printed single-sided (children put pieces on the pages): K-1, 8 pages, 4 copies (3 children + 1 spare); grades 2-3, 9 pages, 5 copies; grades 4-5, 10 pages, 4 copies. Three copies of this guide, and one copy of the share page (section 10).
- Pieces. K-1: one checkers set (24 pieces) or large counters. At most twelve are in play at once on the strips (four per child) and eight on board P, which the whole table plays on one page. The K-1 cells are 1 in square (1.15 in on boards M and N); standard 1-inch checkers fit, 1.25-inch ones overhang a little, which is fine, and pennies work as a backup. Grades 2-3 and 4-5 use pennies (cells are 0.8 in, 0.72 in on board U): a cup of 50 per table covers everything, including the twenty-peg board U at grades 4-5 and a 2-3 table that borrows it. 150 pennies in all, sorted into three cups beforehand.
- One pencil per child. A few blank printed grids (any size) at each table for children who want to invent their own boards.
- For the launch: a whiteboard or large sheet on which a strip of five squares can be drawn, and three magnets or large counters; or simply hold up a K-1 page 1 with checkers on it.

- Preparation, about fifteen minutes: print and sort; each adult plays through the first two problems of their table with real pieces; the K–1 adult also plays boards M, N and P (answers in the box at the start of section 5) so the jumps are familiar; the organizer puts the share page where it can be grabbed at 0:56.

3. The whole-group launch (two to three minutes)

Children stay at their tables; every child has page 1 and pieces. Organizer:

“Find strip A on your first page and put a piece on each gray square. Here is the only move we use today.” (Show it on the board: three squares, pieces on the first two.) “A piece jumps over the piece right next to it and lands on the empty square just beyond. The piece it jumped over comes off. Do it now on your strip. Any jump is fine for now; the grades 4–5 strip has a thick line on it, and you may jump across it or not.” Wait until every child has jumped and removed a piece; table adults help.

“Three things are not allowed: jumping over an empty square, landing on a square that already has a piece, and jumping slantwise. Today’s question is this: if a bunch of pieces start on one side of a line, how far past the line can they send one piece? Your table has its own way of finding out, and you will be working with the people next to you, not just on your own.” Start the tables. Do not say anything about answers or about the golden ratio; that story belongs to the share.

4. Running a table

Every sequence on the pages is *act, record, compare, explain*. Let the children do and record before you ask why. When a child says “it is impossible”, the question to ask, once the recording space is filled, is “How do you know for sure?” Accept explanations a child can produce: pointing at a filled table, listing every case, describing a strategy, showing one counterexample, or saying why the same reason works again on a bigger strip. Do not ask for a rule with a letter in it.

Two habits to enforce gently all hour: the jumped piece really comes off, and pieces are reset to the gray squares before a new try. When a problem asks for “the farthest square” or “the highest row”, the record is the best result over all tries, not the last try.

Working together. The pages build in four ways of working with the other children, and the adult’s job is to keep them honest:

- *Pairs on one board* (K–1 Problems 3–4, grades 2–3 Problems 3–4, grades 4–5 Problem 2): one child plays, the other counts or records, and they swap after each strip. At K–1 the first grader can write the numbers for a kindergartner; the adult writes for anyone who asks. With three children, the third child checks that the jumped piece came off.
- *Design for your neighbour* (K–1 Problem 5, grades 2–3 Problem 5 and Extra 13, grades 4–5 Problem 2 strip E): a child draws a starting position, passes the page, and the neighbour plays it. The designer may not say what they expect; afterwards the two compare what happened with what the designer thought would happen.
- *Table jobs* (the shape hunts: grades 2–3 Problem 11, grades 4–5 Problem 8; K–1 Extra 10 in a small way): the shapes are shared out, every child reports to the table, and the result goes in the table-results row on the page. A shape reported as working is replayed by a different child before the table writes yes. Insist on this; it is the children’s first taste of checking a claim.
- *One-board puzzles* (the T of eight: K–1 Extra 11, grades 2–3 Problem 12, grades 4–5 Problem 12; the twenty pegs: grades 4–5 Extra 16): one board in the middle of the table, pennies on it, the table agrees on each jump before it is made, and the children take turns writing the jumps on the lines (at K–1 the adult makes the jumps and keeps the list). Four children with four private boards is not the point.

At the share, a child from a different table says each result back (“What did the 2–3 table find?”) before the next table shows its own.

5. Table K–1 (two kindergartners and a first grader; adult reads aloud and scribes)

Answer box. Everything the table needs, in one place; the details follow.

Problem 1	Strip A (checkers on squares 2, 3): 2 jumps over 3 and lands on 4, or 3 over 2 lands on 1. Strip B (3, 4): the leftward jump is 4 over 3, landing on 2.
Problem 2	Strip C: NO (no checker is next to another). Strip D: NO (no empty square to land on). Strip E: YES, and only the middle checker can jump.
Problem 3	Strip F (2, 3): always 1 left. Strip G (2, 3, 5): 1 or 2, order matters. Strip H (1, 2, 4, 6): 1 or 2, order matters. Both G and H can be brought down to 1.
Problem 4	Strip I (3, 4): 3 over 4 lands on square 1 across the river. Strip J (1, 2, 3, 4): the play is forced (3 over 4 across, then 1 over 2 to 3, stuck); farthest square 1.
Problem 5	Strip K: whatever dots were drawn, the farthest square is 1. Circle 1.
Problem 6	Board L: C1 jumps over C2 and lands on C3, row 1 across, directly above.
Problem 7	Board M (block): a checker reaches row 1 across (C4 or D4), never the star.
Problem 8	Board N (the L): the star in three jumps: C2 over C3 to C4; E3 over D3 to C3; C3 over C4 to C5.
Extra 9	Three checkers never reach the star; the best is row 1 across. Circle NO.
Extra 10	Only the L of board N and its mirror image (two stacked under the star, two more along the river to the <i>left</i>) reach the star.
Extra 11	Board P: C3 over C4 to C5; C1 over C2 to C3; A4 over B4 to C4; C4 over C5 to C6; E4 over D4 to C4; C3 over C4 to C5; C5 over C6 to C7.

Order and timing. Problems 1–2 (0:08–0:16), 3 (0:16–0:26), 4–5 (0:26–0:34), 6–8 (0:34–0:48). Extras 9–11 only if time is left; if the table reaches 0:44 with board N done, go to Extra 11 and play it together. If the table is behind at 0:40, skip to Problem 8 (board N) so that the hour ends with the L reaching the star. Read each problem aloud one line at a time: an action line, wait, then the recording line. Write numbers in boxes for children who ask. Every problem has a real answer and the children should feel that they found it, so hold hints until a child has tried twice.

Problem 1. Strip A (pieces on squares 2, 3): the piece on 2 jumps over 3 and lands on 4, or the piece on 3 jumps over 2 and lands on 1; either is a correct jump, and the jumped piece comes off. Strip B (pieces on 3, 4): the leftward jump is 4 over 3, landing on 2.

Watch for. Sliding instead of jumping; jumping over an empty square; forgetting to remove the jumped piece. Ask: “Which piece jumped? Which piece came off? Where did it land?”

Problem 2. Strip C (pieces on 1, 3 and 5): NO, no piece is right next to another one, even with three pieces on the strip. Strip D (all five full): NO, there is no empty square to land on. Strip E (pieces on 2, 3, 4): YES, but only the middle piece can jump (over 3 to 5, or over 2 to 1); each end piece would have to land on an occupied square. Ask which piece can jump and why the end pieces cannot.

Problem 3. Strips are six squares long, numbered from the left; pairs, one jumps and one counts.

- Strip F (2, 3): always 1 left, whichever of the two jumps is made.
- Strip G (2, 3, 5): 1 or 2, and the order matters. Jump 2 over 3 to 4, then 4 over 5 to 6 (or 5 over 4 to 3): one left. But if the first jump is 3 over 2 to 1, the pieces on 1 and 5 are stuck: two left. Three lines of play in all.
- Strip H (1, 2, 4, 6): 1 or 2. The only first jump is 1 over 2 to 3. Then 3 over 4 to 5 followed by 6 over 5 to 4 leaves one piece; but 4 over 3 to 2 instead leaves pieces on 2 and 6, stuck.

After both boxes are filled: “Can you make it 1 on strip G? On strip H?” and let the pair try again.

Watch for. Children who stop while a jump is still possible. Ask “Is any piece next to another piece with an empty

square beyond?”

Hints, in order. “Try starting with the other piece.” “Which jump left two pieces far apart? Can you avoid it?” For strip H after the forced first jump: “Now you have pieces on 3, 4, 6. Two jumps are possible. Try both.”

Problem 4. Strips have four home squares (1–4), the river, then square 1 and square 2 across. Strip I (pieces on 3, 4): 3 jumps over 4 and lands on square 1 across; X there. Strip J (1, 2, 3, 4): the play is forced. The only possible first jump is 3 over 4 onto square 1 across; the only possible second jump is 1 over 2 to 3; then the pieces on home 3 and on square 1 across are two apart and stuck. Farthest: square 1.

Watch for. A child who wants to jump from square 1 across back over home square 4 (it is empty after the crossing, so that is not a jump).

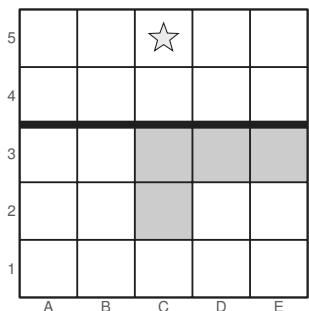
Problem 5. Strip K: whatever squares the neighbour chose, no piece ever reaches the star; every choice of home squares was checked by computer, and the proof for strips of any length is in section 6, Problem 9, and section 9. Circle 1. At this table it is enough to compare the strips and to notice, with the child, what a crossing does: the moment a piece lands on square 1 across, home squares 3 and 4 are empty, and to land on the star a piece would need home square 4 full *and* square 1 across full at the same moment. A way to say it to a child: “refilling square 4 is itself a jump, which uses up two more pieces behind it and leaves two more empty squares.” That is a picture, not a complete check; the complete check at this table is the forced line of strip J (Problem 4) together with letting the child exhaust the other placements, and the complete proof is the values argument in section 6.

Hints, in order. “What do the two squares just before the river look like right after a crossing?” “What would a piece need in order to land on the star?”

Problem 6. Board L: the piece on C1 jumps over C2 and lands on C3, in row 1 across, directly above. X there. (The letters under the boards are for this guide; the children can ignore them.)

Problem 7. Board M (the two-by-two block on C2, D2, C3, D3): the highest square reached is in row 1 across, never the star. In the natural line each column sends a piece up once, the two crossed pieces then sit side by side in row 1, and they can only jump sideways along that row. Other lines start with a sideways jump inside the block; in every line the highest square reached is in row 1 across. Let the children exhaust it.

Problem 8. Board N (the L: two pieces stacked under the star’s column, two more along the river to the right). Three jumps reach the star: the lower stacked piece jumps up over the upper one into row 1; the far right piece jumps left over its neighbor into the square just vacated below the river; that piece jumps up over the row-1 piece onto the star.



Jumps on board N: C2 over C3 to C4; E3 over D3 to C3; C3 over C4 to C5 (the star).

Hints, in order. “Which piece will be the last to move, and what must be right under the star just before?” “Get one piece across first. Now the square under it is empty; can something jump into it from the side?”

Extra 9. Three pieces never reach the star; the best is row 1 across (two stacked pieces cross, the third cannot help). Circle NO. Complete reason, which a first grader can follow with the pieces: to land on the star you need, at one moment, a piece in row 1 under the star and a piece on the home square directly below it. Making the row-1 piece uses two pieces and empties the home square below it; the third piece is not on that square (the crossing just emptied it) and cannot jump into it, because a jump needs a second piece to hop over. (Section 7, Problem 11 gives the values version: three home squares are worth at most 18 and the star is worth 21.) The adult draws each child’s placement on small board 9 and marks the highest square with an X.

Extra 10. Among all ways of putting four pieces on the fifteen home squares of this 5-wide board, exactly two reach the star: the L of board N and its mirror image (two stacked under the star, two more along the river to the *left*). This was checked by computer for all 1365 placements. Every other four-piece shape stops in row 1 or lower. The block never crosses twice in one column; a row of four along the river cannot cross at all (crossing needs two pieces stacked); a T shape crosses once and is stuck. Draw each child's shape on small board 10, put the three small boards side by side, and ask what the shapes that reached the star have in common.

Hints, in order. "Where were the two stacked pieces on board N? Try putting the other two on the other side."

Extra 11. Board P, eight pieces in a T, played by the whole table on one page: five along the river (call them left end, left, middle, right, right end) and three under the middle one (upper, middle, lower). The children call the jumps and the adult makes them. Seven jumps reach the big star in row 3:

1. the upper piece of the column jumps up over the middle river piece into row 1;
2. the lower piece of the column jumps up over the column's middle piece;
3. the left end piece jumps over its neighbor into the middle river square;
4. that middle river piece jumps up over the row-1 piece into row 2;
5. the right end piece jumps over its neighbor into the middle river square;
6. the piece just under it jumps up over it into row 1;
7. the row-1 piece jumps over the row-2 piece onto the big star.

In the letters printed under the board (columns A–E, rows 1–7, star at C7): C3 over C4 to C5; C1 over C2 to C3; A4 over B4 to C4; C4 over C5 to C6; E4 over D4 to C4; C3 over C4 to C5; C5 over C6 to C7. Do it once yourself beforehand. If the children reach only row 2, that is a real result; show the last three jumps and let them call the whole thing again from the start.

If the table stalls. Reading is the bottleneck, not the mathematics: shorten by reading only the action lines. If Problem 3 drags, do strips F and G only and go to the river (Problem 4), which is the heart of the hour. If a child is done listening, hand them a blank grid and ask for "a strip pattern of four pieces that clears to one piece" (for example pieces on squares 1, 2, 4, 6; test any answer by playing it).

If the table finishes early. Extras 9–11 first. Then: (a) Comb strips: draw a long strip and put pieces on squares 1, 2, 4, 6, 8, 10, . . . ; it can always be cleared to one piece: jump 1 over 2 to 3, so that 3 and 4 form a pair; jump 3 over 4 to 5, so that 5 and 6 form a pair; and so on to the last tooth, where the final pair collapses to one piece. Ask "add one more tooth, does it still work?" (b) A tall column two squares wide: no matter how deep, no piece ever gets past row 1, because with only two columns no sideways jump is possible, so each column is a strip on its own. Let them try a 2-by-4 column. (c) Ten pieces, five along the river and five more under them (five wide, two deep): a piece reaches row 3, above the middle column; this is the nicest early-finish target, and it works on the 5-wide board of the packet or on any wider grid. A three-by-three block of nine pieces reaches row 2, but only above a side column of the block, never above its middle column; a three-wide block four deep (twelve pieces) reaches row 3 above a side column. All three facts were checked by computer on the 5-wide board and on a wider one; let them search.

6. Table grades 2–3 (four third graders)

Order and timing. Problems 1–3 (0:08–0:20), 4–5 (0:20–0:27), 6–9 (0:27–0:37), 10–11 (0:37–0:45), 12 (0:45–0:50). Problem 12, the T of eight on one board, is the finale: if the table is behind at 0:42, go to it from wherever you are. Extras 13–16 are for a table that finishes; most tables will not reach them, and that is fine.

Problem 1. Strip A (pegs on 3, 4): 3 over 4 lands on 5, or 4 over 3 lands on 2.

Problem 2. Strip B (3 and 5): not possible, no two pegs are adjacent. Strip C (all eight): not possible, no empty landing square. Strip D (3, 4, 5): possible; only the middle peg can jump, landing on 2 or on 6, so X on squares 2 and 6.

Problem 3. Eight-square strips, numbered from the left; pairs, one jumps and one records, swapping after each strip.

- Strip E (2, 3, 5): 1 is possible (2 over 3 to 4, then 4 over 5 to 6 or 5 over 4 to 3); starting with 3 over 2 to 1 strands pegs on 1 and 5, leaving 2. Fewest: 1.
- Strip F (3, 4, 5): always 2 (two lines of play, both stuck after one jump).
- Strip G (1, 2, 4, 6): 1 is possible: 1 over 2 to 3, 3 over 4 to 5, then 6 over 5 to 4 (or 5 over 6 to 7). Choosing 4 over 3 to 2 as the second jump leaves 2 and 6 stuck. Fewest: 1.
- Strip H (2–6, five in a row): always 3; four lines of play, every one stuck with three pegs, no two of them adjacent.

A certificate the mathematician can use for strip F, after the children have exhausted it: color the squares 1, 2, 3, 1, 2, 3, ... from the left. Every jump changes the number of pegs on *each* of the three colors by one (two colors lose a peg, the third gains one), so the three parities all flip together. A single peg has one odd color and two even ones. Strip F starts with one peg of each color (all odd); flipping all three parities never produces one odd and two even, so one peg is unreachable. Strip H starts with parities one odd, two even, like a single peg, and still gets stuck at 3: the certificate can prove impossibility but never possibility.

Problem 4. Six home squares, the river, then squares 1 and 2 across. Strip I (5, 6): 5 over 6 lands on square 1 across. Strip J (4, 5, 6): only 5 can jump, onto square 1 across (or back to 3); farthest 1. Strip K (3–6): two first jumps are possible (5 over 6 across, or 4 over 3 back to 2), four lines of play in all, every one ending with the farthest square 1. Strip L (1–6): the play is forced: 5 over 6 onto square 1 across, then 3 over 4 to 5, then 1 over 2 to 3; pegs on 3, 5 and square 1 across, all two apart, stuck. Farthest 1.

Problem 5. Strip M, designed by a neighbour: any placement of up to six pegs ends with the farthest square 1. (Computer check over every subset of the six home squares; the proof for strips of any length is Problem 9 and section 9.) After the swap, ask the designer what they expected and whether the player found anything the designer did not.

Problem 6. Values on strip N: 1, 1, 2, 3, 5, 8 on the home squares, 13 on square 1, 21 on square 2.

pegs on	before	after	
all six home squares; the only jump (5 over 8 onto 13)	20	20	same
2 and 3; left (3 over 2 onto 1)	5	1	down
2 and 3; right (2 over 3 onto 5)	5	5	same
8 and 13; left (13 over 8 onto 5)	21	5	down

Ask, after the table is filled: “When did the total stay the same? When did it drop? Did it ever go up?” Do not give the reason yet; Problems 7 and 8 are where the children find it.

Problem 7. No jump on strip N makes the total go up; the children’s three tries will all say *same* (every jump to the right) or *down* (every jump to the left), and the answer to circle is *no*. Let them hunt for a while: a jump to the left from far down the strip (13 over 8 onto 5) is the most tempting, and it drops. If a pair claims an *up*, have them replay it with the pennies; the usual slip is forgetting that the jumped peg comes off.

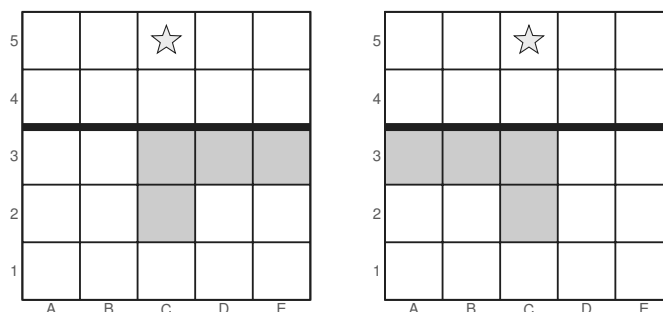
Problem 8. Any jump to the right as a sum: $1 + 1 = 2$, $1 + 2 = 3$, $2 + 3 = 5$, $3 + 5 = 8$, $5 + 8 = 13$ or $8 + 13 = 21$. The reason: a jump to the right removes two neighboring values and lands on the next value, and each value was built to be exactly the sum of the two before it, so the total loses $a + b$ and gains $a + b$. A jump to the left removes a value c and its left neighbor b and lands on the value a before them; since $a + b = c$, the total loses $b + c$ and gains only a , so it drops (by $2b$). This is the whole mechanism; make sure each child can say the right-jump half in their own words, pointing at the strip.

Problem 9. Total of the home squares 20; square 2 is worth 21. The argument, in the children’s terms: the total starts at 20, or less if fewer pegs are used; no jump ever makes the total bigger (Problem 7); if a peg ever stood on square 2 the total would be at least 21; so it never does. Check that every child can say why “at least 21” (the star alone is 21 and all other values are positive) and why “or less” (every value is positive, so leaving pegs out only

lowers the start). This is a complete proof for strip N, whatever the starting squares, and it settles Problems 4 and 5 at once.

Problem 10. Board O: C2 over C3 lands on C4.

Problem 11. The shape hunt, shared out around the table. Only shapes 5 and 6 reach the star, and they are mirror images of each other:



Shape 5 (C3, C2, D3, E3): C2 over C3 to C4; E3 over D3 to C3; C3 over C4 to C5. Shape 6 (A3, B3, C3, C2): C2 over C3 to C4; A3 over B3 to C3; C3 over C4 to C5. Shape 1 (the block C3, D3, C2, D2) reaches row 4 only; shape 2 (the row B3–E3 along the river) cannot cross at all, since crossing needs two pegs stacked; shape 3 (the T: B3, C3, D3, C2) crosses once and is stuck in row 4; shape 4 (two stacks, B and D) sends up two pegs that land two apart in row 4 and cannot interact. Among all 1365 ways to place four pegs on the fifteen home squares of this 5-wide board, only shapes 5 and 6 reach the star (computer check). Insist on the replay rule before the table writes yes. If a child asks why three pegs never work, the complete reason is in section 7, Problem 11 (three home squares are worth at most 18, the star 21); show them board P from the grades 4–5 packet if you like, but it is not part of this table’s sequence.

Hints, in order. “Which square must hold a peg right before the last jump, and which one below it?” “After a peg crosses, refill the square below it from the side.”

Problem 12. Board P (T of eight), one board for the table: C3 over C4 to C5; C1 over C2 to C3; A4 over B4 to C4; C4 over C5 to C6; E4 over D4 to C4; C3 over C4 to C5; C5 over C6 to C7. Values (section 7, Problem 13): the eight squares are worth exactly 55, the star’s value, so every jump in a solution must keep the total and the solution ends with a single peg.

Hints, in order. “What must be on C6 and C5 just before the end?” “Get a peg to C5 first, then refill C4 from the side, then jump C4 over C5.” “Now C5 is empty again and C6 is full: you need another peg on C5 with a peg on C4 under it.”

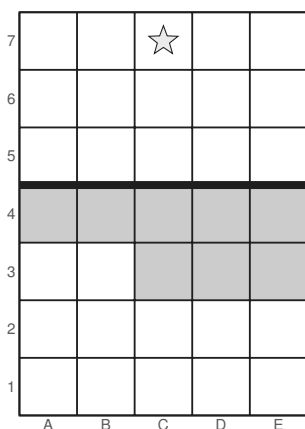
Extra 13. Five-peg patterns on eight squares that clear to one peg (all ten of them, positions from the left): 1-2-3-4-6, 1-2-4-6-8, 1-2-4-7-8, 1-2-5-7-8, 1-3-4-5-6, 1-3-5-7-8, 2-3-4-5-7, 2-4-5-6-7, 3-4-5-6-8, 3-5-6-7-8. Patterns that always end with three or more: five in a row anywhere (3 left), 1-2-3-6-8 (4 left), 1-3-6-7-8 (4 left), and many others; a pattern passed to a neighbour is checked by playing it out, and “cannot be done” means every order was tried.

Extra 14. Strip S: 1, 1, 2, 3, 5, 8, 13 on the seven home squares, 21 on square 1, 34 on square 2. Home total 33, square 2 is 34. Same reason as Problem 9. The children may notice that the home total is always exactly one less than the star (12 and 13 on a strip with five home squares; 20 and 21; 33 and 34); this is a property of these numbers (section 9) and worth saying aloud as “the army is always one short”.

Extra 15. Three pegs never reach C5; the highest row any peg reaches is 4 (for instance from C2, C3 and anything). The reason in pegs: the last jump needs C4 and C3 full at the same moment; filling C4 uses two pegs and empties C3; the third peg cannot refill C3 alone, because a jump needs a second peg to hop over. The values reason is section 7, Problem 11.

Extra 16. The only other eight-peg sets on board U that reach C7 are the two “flag” shapes: A4 B4 C4 D4 E4 with C3 D3 E3, and its mirror image A4 B4 C4 D4 E4 with A3 B3 C3 (checked by computer: any eight-square set worth

at least 55 must contain C4, B4, D4, C3, and then either four of the 5-valued squares, none of which works, or three of them and one of the five 3-valued home squares of the board, giving 50 sets of which exactly the T and these two work).



Right flag: E3 over E4 to E5; D3 over D4 to D5; E5 over D5 to C5; C4 over C5 to C6; A4 over B4 to C4; C3 over C4 to C5; C5 over C6 to C7.

Left flag (mirror image): C3 over C4 to C5; E4 over D4 to C4; C4 over C5 to C6; B3 over B4 to B5; A3 over A4 to A5; A5 over B5 to C5; C5 over C6 to C7.

If the table stalls. Problem 3 can absorb a whole hour; after twelve minutes move on even if strip H is unfinished. If the values table (Problem 6) meets resistance, do the rows together with pennies on the strip, saying each sum aloud (“5 and 8 go away, 13 arrives”). At 0:42, go to Problem 12 whatever else is unfinished; if Problem 12 itself is too long, give jumps 1–3 and let them find 4–7.

If the table finishes early. Extras 13–16 in that order; Extra 16 is the long one. Then: (a) the values on a strip with nine home squares and whether the army is again one short (1, 1, 2, 3, 5, 8, 13, 21, 34 sum to 88; square 2 is worth 89). (b) Board U from the grades 4–5 packet (twenty pegs, star in row 4; solution in section 7, Extra 16). (c) Blocks, on the 5-wide board or any wider one: a two-wide column of any depth never passes row 1 (each column is a strip; no sideways jump exists); a three-by-three block reaches row 2 above a side column; five pegs along the river with five more under them (five wide, two deep) reach row 3 above the middle column; a five-by-four block of twenty pegs reaches only row 3. All checked by computer.

7. Table grades 4–5 (two fourth graders and a fifth grader)

Order and timing. Problems 1–2 (0:08–0:13), 3–5 (0:13–0:24), 6–7 (0:24–0:30), 8 (0:30–0:37), 9–11 (0:37–0:43), 12–13 (0:43–0:50). The values on the strip (3–6) are the heart of this table and deserve their eleven minutes; the T of eight (12) is the finale. If the table is behind at 0:40, go to Problem 12 and come back to 9–11 only if time remains. Extras 14–16 are for a group that finishes; Extra 16 (row four, nineteen jumps) is a genuine puzzle and the answer is below.

Problem 1. Strip A (pegs on 5, 6 of the six home squares): 5 over 6 lands on square 1 across.

Problem 2. Strip B (4, 5, 6): only 5 can jump; farthest square 1. Strip C (3–6): two first jumps, four lines of play, each ending with the farthest square 1. Strip D (1–6): forced play, 5 over 6 across, 3 over 4 to 5, 1 over 2 to 3, stuck; farthest 1. Strip E, designed by a neighbour: whatever the placement, 1. (Every subset of the six home squares was checked by computer; the proof for any strip is Problems 3–6 and section 9.)

Problem 3. Values 1, 1, 2, 3, 5, 8 on the home squares, 13 on square 1, 21 on square 2.

pegs on	before	after	
all six home squares; the only jump (5 over 8 onto 13)	20	20	same
2 and 3; left (3 over 2 onto 1)	5	1	down
2 and 3; right (2 over 3 onto 5)	5	5	same
the second 1 and 2; left (2 over 1 onto the first 1)	3	1	down
8 and 13; left (13 over 8 onto 5)	21	5	down

Ask “When did it stay the same, when did it drop, did it ever go up?” and leave the reason to Problem 4.

Problem 4. There is no jump on strip F that makes the total go up: every jump to the right keeps it and every jump to the left lowers it, so the three recorded tries say *same* or *down* and the answer to circle is *no*. The reason, which the children should produce: a jump to the right removes two neighboring values and lands on the next one, which is their sum by the way the values were built, so the total loses $a + b$ and gains $a + b$; a jump to the left removes c and its left neighbor b and lands on a with $a + b = c$, losing $b + c$ and gaining a , a drop of $2b$. (A jump to the left from the second 1 is impossible, as it would land off the strip.) A claimed *up* is replayed with pennies; the usual slip is forgetting to remove the jumped peg.

Problem 5. Home total 20, square 2 worth 21. Proof for strip F: the total starts at 20, or less with fewer pegs, since every value is positive; no jump raises it (Problem 4); a peg on square 2 would make the total at least 21, because the star alone is 21 and the other pegs add positive amounts. So no peg ever lands on square 2, from any starting set. This settles Problem 2 for every placement at once, which is worth saying aloud.

Problem 6. Strip G: the values are forced, square by square. With 1 and 1 on the two leftmost squares, the jump 1 over 1 must land on a square worth $1 + 1 = 2$; then the jump from 1 over 2 must land on 3; and so on: 1, 1, 2, 3, 5, 8, 13 on the seven home squares, 21 on square 1 and 34 on square 2. (A child who writes a different third value finds that the jump of 1 over 1 onto that square then changes the total; the only freedom is the two starting 1s, and doubling them doubles everything.) Checking jumps away from the star: 13 over 8 onto 5 loses 21 and gains 5, and similarly all the way down. Home total 33, square 2 worth 34; the same argument as Problem 5, word for word. Five home squares: $1 + 1 + 2 + 3 + 5 = 12$ against 13; eight: $1 + 1 + 2 + 3 + 5 + 8 + 13 + 21 = 54$ against 55. Why always one short: when the strip grows by one home square, the old square 1 becomes a home square, so the army’s total grows by the old square-1 value; and the new star is the old star plus the old square 1 (the next value is the sum of the two before it), so the star grows by exactly the same amount. The difference therefore never changes, and it is 1 at the start (two home squares worth 1 and 1 against a star of 3). This is an induction, and a fifth grader can say it with the pennies. Why a strip of any length: the values can be continued as far as needed, and the army is one short every time.

Problem 7. Board O: C2 over C3 lands on C4.

Problem 8. The shape hunt, two shapes per child. Only shape L (the mirrored L: A3, B3, C3, C2) reaches the star: C2 over C3 to C4; A3 over B3 to C3; C3 over C4 to C5.

5			☆		
4					
3					
2					
1					
	A	B	C	D	E

Shape I (the block C3, D3, C2, D2) reaches row 4 only; shape J (the Z: B2, C2, C3, D3) crosses once and is stuck in row 4; shape K (the column C1, C2, C3 with D3) crosses and the rest cannot refill C3; shape M (the S: C3, D3, D2, E2) and shape N (the tall L: C1, C2, C3, D1) reach row 4 at best. Among all 1365 placements of four pegs on the fifteen home squares of this 5-wide board, only shape L and its mirror image (the L to the right, which the 2–3 table has) reach the star (computer check). Insist on the replay before *yes* goes in the table results.

Hints, in order. “Which two squares must both be full just before the last jump?” “Once a peg has crossed, the square under it is empty; refill it from the side.”

Problem 9. Board P values (rows 5 to 1): 8 13 21 13 8 / 5 8 13 8 5 / 3 5 8 5 3 / 2 3 5 3 2 / 1 2 3 2 1. (a) C2 over C3 to C4: 13 before, 13 after. (b) E3 over D3 to C3: 8 and 8. (c) B3 over C3 to D3: 13 before, 5 after. (d) C4 over C3 to C2: 21 before, 5 after.

Problem 10. The rule: count a square's distance from the star in steps along rows and columns; 0 steps is worth 21, 1 step 13, 2 steps 8, 3 steps 5, 4 steps 3, 5 steps 2, 6 steps 1, each value the sum of the next two further out (the Fibonacci numbers, read backward from the star). No jump on board P makes the total go up, and there are exactly three kinds of jump to check: toward the star along a row or column (the landing square is one step nearer than the jumped square, which is one step nearer than the start, so the two removed values are the two before the landing value and the total is unchanged, as in (a) and (b)); away from the star (the mirror image, and the total drops, as in (d)); and sideways past the star's row or column, where the jumped square is nearer than both the start and the landing square (B3 over C3 to D3: the total drops by the value of the jumped square, as in (c)). The children should find all three kinds in their search and see that none goes up.

Problem 11. Best three home squares $8 + 5 + 5 = 18$ (C3, B3, D3); star 21. With pennies on exactly those three squares the only jumps are sideways along the river (C3 over D3 to E3 or C3 over B3 to A3), so the highest row reached is 3: the most valuable three cannot even cross. Shape I (block) $8 + 5 + 5 + 3 = 21$; shape L $3 + 5 + 8 + 5 = 21$; shape M $8 + 5 + 3 + 2 = 18$. So three pegs, placed anywhere, are worth at most 18 and can never reach a square worth 21: the question of three pegs is settled for all placements at once, without trying them. Shape M is worth 18 and fails for the same reason. Shape I is worth 21 and still fails, so enough value does not guarantee success; the values only ever prove impossibility. Shape L spends its 21 exactly, each jump keeping the total, and ends with one peg on the star.

Problem 12. Board Q, one board for the table: C3 over C4 to C5; C1 over C2 to C3; A4 over B4 to C4; C4 over C5 to C6; E4 over D4 to C4; C3 over C4 to C5; C5 over C6 to C7. A group that plans backward ("C7 needs C6 and C5 full; C6 needs C5 and C4 full . . .") finds it in a few minutes; a group jumping at random may need the first hint.

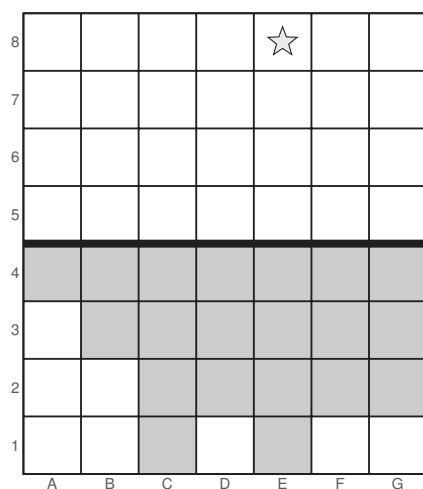
Hints, in order. "Plan the last jump first." "The column of three under the middle is there to fill C5 twice." "The two ends of the row of five are there to fill C4 twice."

Problem 13. Board R values (rows 7 to 1): 21 34 55 34 21 / 13 21 34 21 13 / 8 13 21 13 8 / 5 8 13 8 5 / 3 5 8 5 3 / 2 3 5 3 2 / 1 2 3 2 1. Seven most valuable home squares: $13 + 8 + 8 + 8 + 5 + 5 + 5 = 52$; star 55; the T of board Q: $13 + 8 + 8 + 5 + 5 + 8 + 5 + 3 = 55$. So seven pegs never reach C7 (52 is less than 55, and the total never rises, by the same three-kinds check as on board P), and any eight-peg army that does must be worth at least 55 and spend it exactly: every jump keeps the total and one peg is left.

Extra 14. Three pegs never reach C5; the highest row reached is 4 (e.g. from C2, C3 and any third square). Full reason: Problem 11.

Extra 15. The only other eight-peg sets that reach C7 are the two flags: A4 B4 C4 D4 E4 with C3 D3 E3 (E3 over E4 to E5; D3 over D4 to D5; E5 over D5 to C5; C4 over C5 to C6; A4 over B4 to C4; C3 over C4 to C5; C5 over C6 to C7) and its mirror image A4 B4 C4 D4 E4 with A3 B3 C3 (C3 over C4 to C5; E4 over D4 to C4; C4 over C5 to C6; B3 over B4 to B5; A3 over A4 to A5; A5 over B5 to C5; C5 over C6 to C7). Why nothing else: an eight-square set worth at least 55 must contain C4 (13) and B4, D4, C3 (8 each), and then either four of the five 5-valued squares (57 in all; none of the five such sets works) or three 5-valued squares and one of the five 3-valued home squares of the board (55 exactly; of these 50 sets, exactly the T and the two flags work). Both statements were checked by exhaustive computer search on this 5-wide board.

Extra 16. Board U (columns A–G, rows 1–8, river above row 4, star at E8). Twenty pegs: A4 B4 C4 D4 E4 F4 G4 (row 4); B3 C3 D3 E3 F3 G3 (row 3); C2 D2 E2 F2 G2 (row 2); C1 E1 (row 1). Nineteen jumps:



1. G3 over G4 to G5
2. F3 over F4 to F5
3. G5 over F5 to E5
4. E4 over E5 to E6
5. E2 over E3 to E4
6. G2 over F2 to E2
7. E1 over E2 to E3
8. E3 over E4 to E5
9. E5 over E6 to E7
10. D3 over D4 to D5
11. C3 over C4 to C5
12. C5 over D5 to E5
13. C1 over C2 to C3
14. B3 over C3 to D3
15. D2 over D3 to D4
16. A4 over B4 to C4
17. C4 over D4 to E4
18. E4 over E5 to E6
19. E6 over E7 to E8

The structure is worth pointing out: jumps 1–9 use only the ten pegs in columns E–G (a three-by-three block plus E1) and park a peg on E7; jumps 10–12 bring a peg from the left half to E5; jumps 13–17 bring a second peg to E4 while E5 is full; jumps 18–19 finish. With Fibonacci values scaled so that the star is 377, the twenty squares are worth exactly 377, so every jump must keep the total. Could nineteen pegs do it? On board U itself the values already settle it: only fifteen home squares lie within seven steps of E8 and six at eight steps, and the nineteen heaviest are worth $55 + 3 \cdot 34 + 5 \cdot 21 + 6 \cdot 13 + 4 \cdot 8 = 372 < 377$. On an unbounded board the nineteen heaviest squares (sixteen within seven steps, three of the nine at eight steps) are worth exactly 377, so the values alone do not exclude them; an exhaustive search over those 84 sets finds that none works, and the literature (section 9) gives twenty as the minimum.

Hints, in order. “The last jump is E6 over E7 to E8, so plan to park a peg on E7 and then bring another to E6 with E5 full.” “The right three columns and the column under E are a row-three army; use them first.” “After the peg is on E7, the left half must deliver a peg to E4 while E5 is full.”

If the table stalls. Problems 3–6 are the heart; give them time and let the children do the sums with pennies on the strip. If Problem 6 stalls at the third value, ask “which square does the jump of 1 over 1 land on, and what must it be worth so that the total stays the same?” If Problem 12 goes past eight minutes, give jumps 1–3. Board U is optional.

If the table finishes early. Extras 14–16 in that order; then: (a) Row five: tell them the true story (section 8) and ask what values would be needed, given that the argument must work for an army of any size; they will see that Fibonacci numbers run out of “room” on an infinite board and the mathematician can introduce the number σ with $\sigma^2 + \sigma = 1$. (b) Narrow armies: a two-wide column of any depth never passes row 1 (no sideways jump is possible, so each column is a strip); a three-wide army of any depth never reaches row 4 in any column (values: with the target above the middle column the three columns are worth $\sigma^2 + 2\sigma^3 = 3\sigma - 1 \approx 0.85 < 1$ in the units of section 9, and less above any other column), though a three-by-four block does reach row 3, above a side column. (c) Blocks, on the 5-wide board or any wider one: a five-by-four block of twenty pegs reaches only row 3 (its value with the star in row 4 above its middle column is 369, less than 377, and less still above any other column), a five-by-five block reaches row 4 (computer check); ask them to find a twenty-peg shape of their own that reaches row 4.

8. The share (eight minutes)

Order: K–1, grades 2–3, grades 4–5, then the organizer. Two minutes each; children hold up pages and pieces, adults keep it moving. After each table, a child from a different table says the result back in one sentence (“What did the K–1 table find?”) before the next table starts.

- K–1 shows strip J (four pieces, the play is forced, the farthest square is 1) and board N (the L shape reaching the star in three jumps).
- Grades 2–3 show strip N with the values: the six pegs are worth 20, square 2 is worth 21, and no jump ever

makes the total bigger. “The army is always one short.” Then the table-results row: two shapes of six reached the star, and they are mirror images.

- Grades 4–5 show the T of eight reaching row 3 and board R: seven pegs are worth at most 52, the star is 55, so eight is the fewest; and, if they got there, board U with twenty pegs reaching row 4.
- Organizer, one minute, holding up the section 10 page: on an endless grid with an endless army below the line, John Conway asked how high a single peg can be sent. Four rows, with twenty pegs, as on this page; and never five, not with any army, however large, in any number of moves. The proof is the same idea the tables used, values that a jump can never increase, with one special number in place of the Fibonacci numbers. Next time we can look at that number.

9. The mathematics behind the session (for the two mathematicians)

The potential. Fix the target square T and give the square at Manhattan distance d from T the weight σ^d , where $\sigma = (\sqrt{5} - 1)/2 \approx 0.618$ satisfies $\sigma^2 + \sigma = 1$. Consider a jump along three collinear squares at distances d_1, d_2, d_3 from T (jumper, jumped, landing). Either the jump moves toward T along that line, $(d+2, d+1, d)$, and the weight change is $\sigma^d - \sigma^{d+1} - \sigma^{d+2} = \sigma^d(1 - \sigma - \sigma^2) = 0$; or it moves away, $(d, d+1, d+2)$, change $\sigma^{d+2} - \sigma^d - \sigma^{d+1} < 0$; or it crosses the line through T , $(d+1, d, d+1)$, change $-\sigma^d < 0$. So the total weight W of the pegs on the board never increases: W is a monovariant. Reaching T needs $W \geq 1$ at that moment, so an initial position with $W < 1$ can never reach T . This is the argument of Berlekamp, Conway and Guy (*Winning Ways*, vol. 4, ch. 23, “Purging Pegs Properly”) and of Aigner’s “Moving into the desert with Fibonacci” (Math. Magazine 70, 1997); Honsberger’s *Mathematical Gems II* also has a chapter on it. (The references in this section were cited from memory and not checked against a library; the organizer may want to confirm volume and page numbers before passing them on.)

Row five is unreachable. With T in row 5, the column under T has weight $\sigma^5 + \sigma^6 + \dots = \sigma^5/(1 - \sigma) = \sigma^5/\sigma^2 = \sigma^3$, and the columns at horizontal distance $k \geq 1$ have weight σ^{3+k} each. The whole half-plane is worth $\sigma^3 + 2 \sum_{k \geq 1} \sigma^{3+k} = \sigma^3 + 2\sigma^4/\sigma^2 = \sigma^3 + 2\sigma^2$. Using $\sigma^2 = 1 - \sigma$ and $\sigma^3 = \sigma - \sigma^2 = 2\sigma - 1$, this is $2\sigma - 1 + 2 - 2\sigma = 1$: the infinite army is worth exactly the target square. A finite army is worth strictly less than 1, and W never increases, so no finite army reaches row five; and any actual play uses finitely many pegs. (For T in row 4 the half-plane is worth $1/\sigma = \phi > 1$, which is why the argument does not exclude row 4, and indeed 20 pegs suffice.)

One dimension. On a strip with T two squares past the line, the home squares are worth $\sigma^2 + \sigma^3 + \dots = 1$ exactly, and one square past the line the whole home side is worth $\sigma^{-1} > 1$. So a finite army reaches square 1 (two pegs do) and never square 2. Whether the army is contiguous or has gaps makes no difference. The Fibonacci version used on the pages is the same argument with the values $F_1, F_2, \dots$ read from the far end of a finite strip: a jump toward T replaces $F_k + F_{k+1}$ by F_{k+2} , and $\sum_{i=1}^n F_i = F_{n+2} - 1$, so a home side of n squares is worth $F_{n+2} - 1$ while square 2 is worth F_{n+2} : “always one short”. The identity is proved by induction (adding F_{n+1} to both sides turns the case n into the case $n+1$; on the page this is “the army and the star grow by the same amount”), and the children can discover it as the pattern 12/13, 20/21, 33/34, 54/55. The grades 4–5 Problem 6 makes the recurrence itself the discovery: once two adjacent values are fixed, “a jump toward the star keeps the total” forces every later value, which is why Fibonacci numbers appear at all. Integer values work on any finite board because all values are positive there; on an infinite board the integer recurrence would have to run backward into 0, 1, −1, . . . , and the geometric σ^d is the right replacement.

Two dimensions on the pages. The value boards use F_{n-d} at distance d with n large enough that every printed square has a positive value (board P: star 21; board R: star 55; for board U take the star to be 377). Everything the pages ask is then exact arithmetic: three home squares are worth at most $8 + 5 + 5 = 18 < 21$, seven at most $13 + 8 + 8 + 8 + 5 + 5 + 5 = 52 < 55$, so 3 and 7 pegs fail for rows 2 and 3, and 4 and 8 succeed by exhibition. The minimal armies for rows 1–4 are 2, 4, 8, 20 (*Winning Ways*; Bell, Hirschberg and Guerrero-García, “The minimum size required of a solitaire army”, INTEGERS 7 (2007), treat this and related boards). For row 4 the potential alone gives 19 as a lower bound on an unbounded board: the nineteen heaviest home squares (the

sixteen within seven steps and three of the nine at eight steps) are worth exactly 1, i.e. 377 in the units of board U, so a nineteen-peg solution would have to be *perfect*, every jump toward the target and the final position a single peg; an exhaustive search over the 84 candidate sets, restricted to weight-preserving jumps, finds none, which is how “20” was confirmed for this guide. On the printed board U, with its seven columns and four home rows, only fifteen home squares lie within seven steps of E8 and six at eight steps, so the heaviest nineteen are worth 372 and the values already exclude nineteen pegs there. The twenty-peg set on board U is worth exactly 377, so its nineteen jumps are all weight-preserving, and it ends with one peg. The exact-equality cases are the subtle part of the whole subject: for row 3 exactly three eight-peg sets reach the star (the T and the two flags), and for row 2 exactly the two L shapes. These counts were checked on the 5-wide boards of the packets and again on wider ones (9 wide for row 2, 7 wide for row 3): the sets of sufficient value are the same there, since the extra squares are worth too little, and the extra room for manoeuvre does not help.

Sufficiency is a different matter. The block of four (worth 21) and the T of four (worth 23) cannot reach row 2; the value argument is a certificate of impossibility only. Deciding reachability in general is hard: solvability of generalized peg solitaire is NP-complete (Uehara and Iwata, 1990), which is why the positive answers on the pages come from explicit jump sequences found by search.

Pagoda functions. In peg solitaire proper (clear a board to one peg), a function p on the squares with $p(a) + p(b) \geq p(c)$ for every jump a over b to c is called a pagoda function; the total of p over the pegs never increases, so a target position worth more than the start is unreachable. Our value boards are pagoda functions for the solitaire-army problem, and the mod-3 coloring used at the grades 2–3 table (three colors, every jump flips the parity of all three counts) is the classical parity certificate of *Winning Ways*; both are necessary conditions only, as strip H and shape I show. Beeler and Hoilman (“Peg solitaire on graphs”, *Discrete Math.* 311, 2011) start the study of the same game on arbitrary graphs.

Facts checked by computer for this session (any of them can be handed to a fast table): all 1365 four-peg placements on the fifteen home squares of the 5-wide row-2 board were searched, and only the two L shapes reach the star (the same on a 9-wide board); all eight-peg sets of sufficient value on the 5-wide row-3 board were searched (T and two flags; the same on a 7-wide board); the 1-D strips of up to eight home squares were searched over every subset; rectangular blocks of pegs with their top row on the line reach at most the following rows, in the best column, on the 5-wide board and on a wider one alike: two wide (any depth tried, and provably any depth) row 1; three by two, three by three, four by two, four by three: row 2; three by four, four by four, five by two, five by three, five by four: row 3; five by five: row 4. A three-wide army of any depth cannot reach row 4 in any column, by the potential: with the target above its middle column its three columns are worth $\sigma^2 + 2\sigma^3 = 3\sigma - 1 \approx 0.854$, above a side column $\sigma^2 + \sigma^3 + \sigma^4 = 2 - 2\sigma \approx 0.764$, and above any other column less still.

Where it leads. The monovariant is the basic tool of olympiad combinatorics (invariants and semi-invariants), and the same idea appears as potential functions in amortized analysis of algorithms and as Lyapunov functions in dynamics: to show a process cannot reach a state, find a quantity that only moves one way. The golden ratio enters because the jump rule forces the recurrence $v_{k+2} = v_k + v_{k+1}$; the Fibonacci strip values, Zeckendorf representations and the continued fraction of ϕ all sit next door. For a next session: the solitaire army on the triangular or hexagonal lattice and in three dimensions (the same potential with a different sum gives different maximal rows), peg solitaire on small graphs (which trees can be solved?), and the reverse question of reconfiguration: which positions can reach which.

10. Page to hold up at the share

Twenty pegs below the line, the star four rows up: this army sends a peg to the star (the nineteen jumps are in section 7, Extra 16). The dashed star five rows up is out of reach for every army, however large, on a grid that goes on forever.

